PI

FRAUD ML & MLOPS · CASE STUDY

Moniguard v2: Nigerian PoD Fraud Scoring Engine

BI Scientist • AI/ML Engineer • Python Developer

Calibrated XGBoost + Isolation Forest ensemble for pay-on-delivery orders. Synthetic data: PR-AUC 0.959 in batch, 0.837 one transaction at a time via the API.

LOGISTICS & FINTECH FRICTION

Why Pay-on-Delivery (PoD) Fraud Is Different

Pay-on-delivery is common in Nigerian e-commerce, and it has its own fraud patterns.

Account & Identity Abuse

- Account takeover on a new device right before a high-value order.
- Recycled phone numbers and identity theft.
- Clusters of orders to the same delivery address.

Refund & Friendly Fraud

- A successful order followed by a refund request.
- Refund abuse and triangulation schemes.
- The merchant needs a score before the rider leaves.

Extreme Imbalance

- Fraud is about 2.8% of transactions in the synthetic data.
- Accuracy is misleading: 97% can mean no fraud caught.
- Report PR-AUC and recall at a fixed alert rate.

ML PIPELINE

73 Features, Chronological Split & Calibrated Ensemble

One feature pipeline shared by training and the FastAPI service.

Strict Time Split

- 70/30 chronological split on 60,000 synthetic transactions.
- Trained on past, tested on future.
- Prevents temporal data leakage.

Hybrid Architecture

- Calibrated XGBoost classifier.
- Isolation Forest for unusual activity.
- Isotonic calibration fit on a held-out slice.

Serving

- FastAPI POST /v1/score, API key required.
- Returns a 0–1000 score, decision and reason codes.
- p95 145 ms end to end, one transaction.

PRODUCT WALKTHROUGH

Moniguard Fraud Workbench

Analyst dashboard showing scored orders and reason codes (synthetic data).

- **Reason Codes:** Each score lists the features that drove it, so merchant ops can act before the rider leaves.
- **Clear Decisions:** Approve, review (score 300 and above) or block (700 and above).
- **MLOps Stack:** Docker Compose with MLflow, Airflow, Postgres, Prometheus and Grafana; a no-Docker demo reproduces the numbers.

XGBoost FastAPI Docker MLflow



github.com/ikebude/MLOps-Fraud-Detection

RESULTS ON SYNTHETIC DATA

Evaluated on 18,000 Later Transactions

From reports/metrics.json (seed 42). The data comes from the repo's own generator, not real transactions.

PR-AUC, batch

0.959

95% CI: 0.942–0.972

PR-AUC, API path

0.837

3,000 scored one at a time

Recall @ 5% Alert

0.994

0.843 on the API path

p95 Latency

61 ms

Model only; 145 ms via the API

Why PR-AUC Matters

With ~2.9% fraud in the test set, ROC-AUC looks better than it is.

PR-AUC measures precision against recall across thresholds.

The batch-to-API gap is the cost of serving without a feature store.

Read Honestly

I wrote the generator, so its fraud is easier than real fraud.

CI retrains twice and fails if the numbers differ.

Next: feature store, then PaySim and IEEE-CIS benchmarks.

OPEN SOURCE & REPRODUCIBLE

Fraud scoring built around Nigerian pay-on-delivery.

Source code, Docker Compose stack, feature pipeline and a one-command reproducible benchmark (make demo).

Author: Precious Ikebude
Repository: github.com/ikebude/MLOps-Fraud-Detection